Enrollment No.: 240840116032

Practical 10 — Mini Project

Hotel Booking Data Analysis

Power BI Dashboard: Hotel Booking Analysis

AIM: Mini Project, analyze a real-world hotel booking dataset, perform data inspection and analysis, create meaningful visualizations, and present useful business insights through a Power BI dashboard.

Project Description

This mini project analyzes a cleaned hotel booking dataset containing **47,883 booking records** and **32 columns**. The dataset includes booking details for City Hotel and Resort Hotel, arrival dates, guest information, market segments, distribution channels, room types, prices, cancellation status, and special requests.

The analysis examines:

- **Booking pattern:** booking counts by year, month, and hotel type.
- **Customer profile:** guest counts, country of origin, and customer type.
- **Booking channels:** market segments and distribution channels.
- **Business performance:** cancellations, average daily rate (ADR), and stay duration.

The cleaned dataset is analyzed in Python and then presented through an interactive Power BI dashboard titled **Hotel Booking Analysis**. The purpose is to understand historical booking patterns and communicate evidence-based business insights.

Phase 1: Load the Dataset

Load the prepared Excel file using Pandas. Keep `cleaned_hotel_bookings.xlsx` in the same folder as this Jupyter Notebook.

In [1]:
```
# Import the required libraries
import pandas as pd
import matplotlib.pyplot as plt

# Load the prepared hotel booking dataset
data = pd.read_excel("cleaned_hotel_bookings.xlsx")

# Display the first five records
data.head()
```

Out [ ]:
```
hotel  is_canceled  lead_time  arrival_date_year  ... required_car_parking_spaces  total_of_special_requests  reservation_status  r
0  Resort Hotel            0        342               2015  ...                           0                          0           Ch
1  Resort Hotel            0        737               2015  ...                           0                          0           Ch
2  Resort Hotel            0          7               2015  ...                           0                          0           Ch
3  Resort Hotel            0         13               2015  ...                           0                          0           Ch
4  Resort Hotel            0         14               2015  ...                           0                          1           Ch
```

Output: The dataset is loaded successfully. It contains 47,883 booking records and 32 columns.

Phase 2: Basic Dataset Inspection

Inspect the dataset before performing analysis. These commands help us understand the number of records, column names, data types, missing values, and statistical distribution.

In [2]:
```
# Display the first five records
data.head()

# Display the last five records
data.tail()

# Check dataset dimensions
data.shape

# Display column names
data.columns

# Check data types and non-null values
data.info()

# Statistical summary of numerical columns
data.describe()

# Check missing values and duplicate records
data.isnull().sum()
data.duplicated().sum()
```

Inspection Item	Result
Total records	47,883

Inspection Item	Result
Total columns	32
Duplicate rows	0
Missing cells	0
Arrival years	2015, 2016, 2017

The inspection confirms the dimensions and quality of the supplied prepared dataset.

Phase 3: Data Cleaning and Feature Engineering

The supplied Excel file is already prepared. The following code documents common cleaning checks and creates two useful features: total guests and total nights.

In [3]:
```
# Create a working copy
df_clean = data.copy()

# Remove exact duplicate rows
df_clean = df_clean.drop_duplicates()

# Fill missing guest counts with zero
for col in ["children", "babies"]:
    df_clean[col] = df_clean[col].fillna(0)

# Fill missing country values
df_clean["country"] = df_clean["country"].fillna("Unknown")

# Fill missing agent and company identifiers with zero
for col in ["agent", "company"]:
    df_clean[col] = df_clean[col].fillna(0)

# Convert selected columns to integer type
for col in ["children", "babies", "agent", "company"]:
    df_clean[col] = df_clean[col].astype(int)

# Create total guests and total nights
df_clean["total_guests"] = (
    df_clean["adults"] + df_clean["children"] + df_clean["babies"]
)
df_clean["total_nights"] = (
    df_clean["stays_in_weekend_nights"] +
    df_clean["stays_in_week_nights"]
)

# Verify the cleaned dataset
df_clean.isnull().sum()
df_clean.shape
```

Cleaning Summary

• Exact duplicate rows found in the supplied file: 0.
• Missing cells found in the supplied file: 0.
• The dataset contains booking, guest, hotel, channel, price, and reservation-status information.
• Two useful derived columns are `total_guests` and `total_nights`.
• ADR has an unusually high maximum compared with its median and should be checked before pricing conclusions are made.

Phase 4: Data Visualization

The following charts summarize the supplied hotel booking data. Each chart can also be recreated in Power BI.

4.1 Bookings by Hotel Type — Bar Chart

Resort Hotel has 33,955 bookings, while City Hotel has 13,928 bookings.



Figure 1: Number of bookings by hotel type.

4.2 Bookings by Arrival Year — Line Chart

This chart compares booking counts across the arrival years available in the dataset.

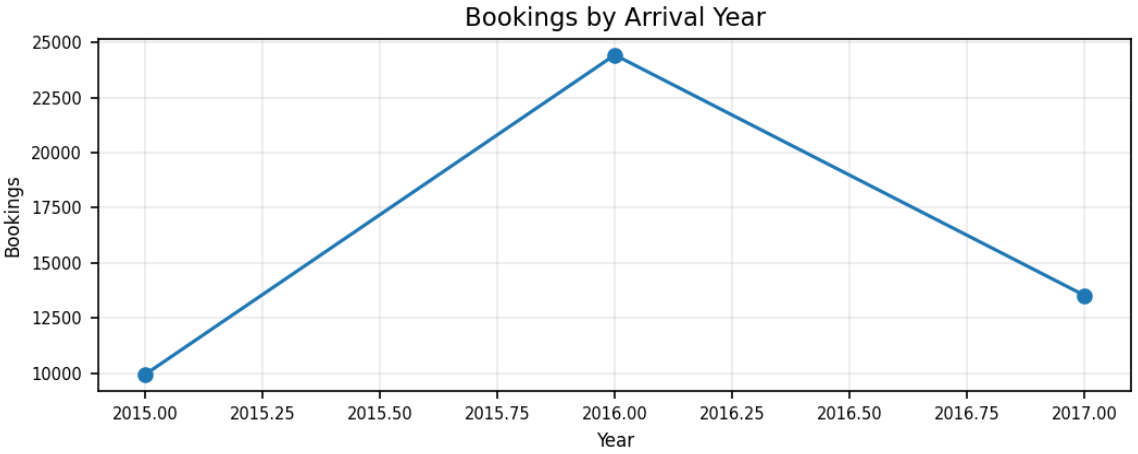


Figure 2: Booking count by arrival year.

4.3 Bookings by Arrival Month — Bar Chart

August has the highest booking count (6,527) in the supplied dataset.

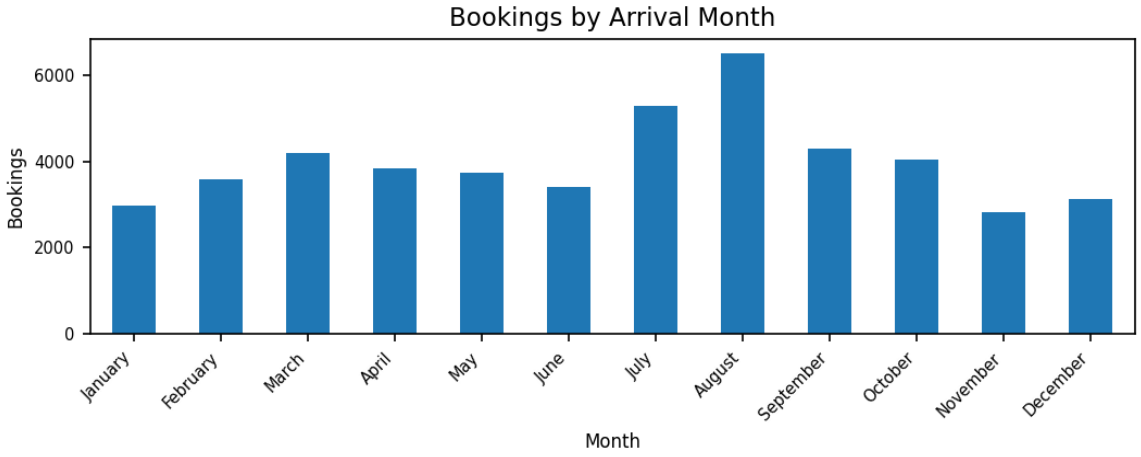


Figure 3: Booking count by arrival month.

4.4 Bookings by Market Segment — Bar Chart

The largest market segment is Online TA with 26,948 bookings.

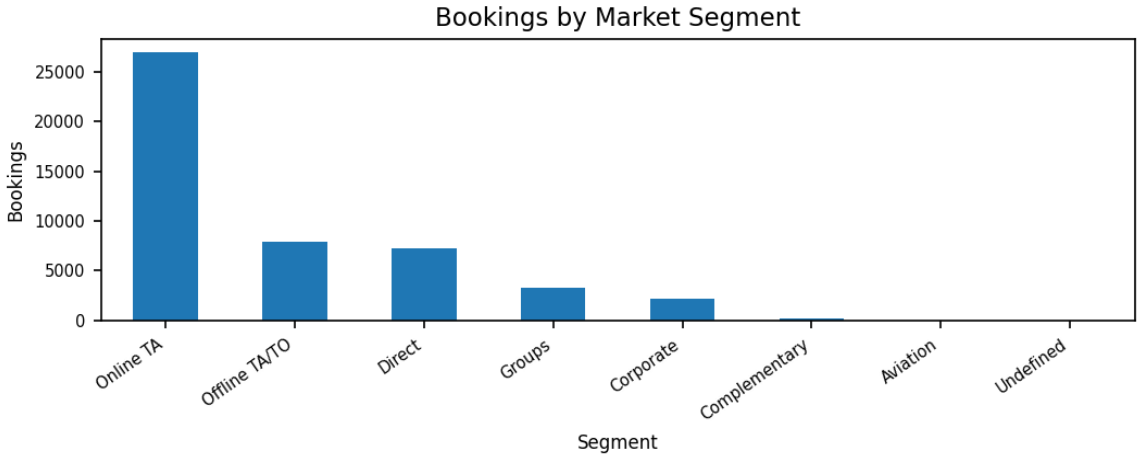


Figure 4: Booking count by market segment.

4.5 Top 10 Countries by Bookings — Bar Chart

The chart shows the ten most frequently recorded country codes in the booking dataset.

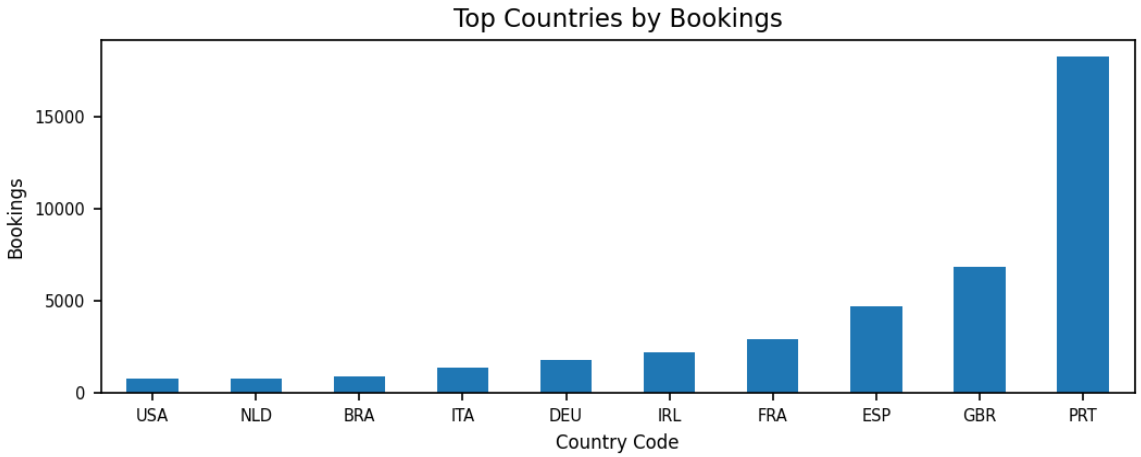


Figure 5: Top ten countries by booking count.

Phase 5: Power BI Dashboard

Import the same cleaned hotel booking dataset into Power BI and create an interactive dashboard titled **Hotel Booking Analysis**.

Suggested dashboard elements:

• **KPI Cards:** Total Bookings, Cancelled Bookings, Cancellation Rate, Average ADR, and Total Guests.
• **Column Chart:** Bookings by Hotel Type.
• **Line Chart:** Bookings by Arrival Year or Month.
• **Bar Chart:** Bookings by Market Segment and Distribution Channel.
• **Map or Bar Chart:** Bookings by Country.
• **Slicers:** Hotel, Arrival Year, Arrival Month, Country, and Customer Type.

Useful Power BI Measures (DAX)

In [4]:
```
Total Bookings = COUNTROWS('HotelBookings')

Cancelled Bookings =
CALCULATE(
    COUNTROWS('HotelBookings'),
    'HotelBookings'[is_canceled] = 1
)

Cancellation Rate =
DIVIDE([Cancelled Bookings], [Total Bookings], 0)

Average ADR = AVERAGE('HotelBookings'[adr])

Total Guests =
SUM('HotelBookings'[adults]) +
SUM('HotelBookings'[children]) +
SUM('HotelBookings'[babies])
```

Replace `'HotelBookings'` with the actual table name shown in your Power BI Fields pane.

Phase 6: Key Insights

• **1. Resort Hotel has more bookings.** Resort Hotel accounts for 33,955 bookings (70.9%), compared with 13,928 City Hotel bookings (29.1%).

• **2. A substantial share of bookings are cancelled.** The dataset records 18,200 cancellations, representing 38.0% of all bookings.

• **3. August is the busiest arrival month.** August has 6,527 bookings, followed by July with 5,301.

• **4. Online travel agencies are an important market segment.** Online TA is the largest market segment with 26,948 bookings.

• **5. TA/TO is the largest distribution channel.** It records 37,180 bookings.

• **6. Portugal is the most frequently recorded country.** PRT appears 18,286 times, followed by GBR (6,865) and ESP (4,683).

• **7. Average Daily Rate (ADR).** The average ADR is 100.47 and the median is 88.00. The maximum value is 5,400.00, which is much higher than the median and should be treated as a possible outlier for further checking.

Ethical Considerations and Data Limitations

• The dataset contains historical booking records and should be interpreted as descriptive information, not as a guarantee of future demand.
• Country codes describe recorded customer origins and should not be used to stereotype customers or communities.
• A booking marked as cancelled should be interpreted according to the dataset's cancellation field and reservation status.
• The unusually high ADR value may affect averages; investigate outliers before making pricing decisions.
• The analysis depends on the coverage, definitions, and accuracy of the supplied dataset.

Conclusion

In this mini project, the hotel booking dataset containing 47,883 records and 32 columns was inspected and analyzed using Python. The analysis examined hotel type, arrival patterns, market segments, distribution channels, customer origins, cancellations, and ADR.

Five visualizations were prepared to communicate booking patterns, and a Power BI dashboard structure with KPI cards, charts, and slicers was proposed. The findings show that Resort Hotel has the larger booking count, cancellations represent a substantial share of records, and August is the busiest arrival month. These results can support further investigation into hotel demand, booking channels, and cancellation management.